

Complete Anaconda + Jupyter Notebook Setup Guide

This guide provides a beginner-friendly setup process for: - Anaconda - Python 3.11 - Jupyter Notebook - Machine Learning Environment - Data Analysis Libraries

Recommended Python Version

Use Python 3.11.

Why?

- Stable and fast
- Compatible with ML libraries
- Works well with Pandas, NumPy, Scikit-learn, TensorFlow, and PyTorch

Avoid:

- Python 3.13
 - Very old versions like 3.8 or 3.9
-

Step 1 — Download Anaconda

Go to:

<https://www.anaconda.com/download>

Download:

- Windows 64-bit Graphical Installer
 - Python 3.11 Version
-

Step 2 — Install Anaconda

Open installer and click:

- Next
- I Agree

Important:

- Check: Register Anaconda as default Python
- Do NOT check: Add Anaconda to PATH

Install location:

C:\Users\YourName\anaconda3

Step 3 — Verify Installation

Open CMD and run:

```
conda --version
```

If installed correctly, conda version will appear.

Step 4 — Fix PATH Issue (Only if Needed)

If CMD shows "conda not recognized", add these paths manually:

```
C:\Users\YourName\anaconda3  
C:\Users\YourName\anaconda3\Scripts  
C:\Users\YourName\anaconda3\Library\bin
```

Steps:

- Open Edit Environment Variables
 - Open Path settings
 - Add paths above
 - Restart CMD
-

Step 5 — Create ML Environment

Run:

```
conda create -n ml_env python=3.11
```

Press y to continue.

Step 6 — Activate Environment

Run:

```
conda activate ml_env
```

If activated successfully, CMD will show:

```
(ml_env)
```

Step 7 — Install Jupyter Notebook

Run:

```
pip install notebook
```

Step 8 — Install Important Libraries

Run:

```
pip install numpy pandas matplotlib seaborn scikit-learn
```

Optional:

```
pip install tensorflow torch
```

Step 9 — Verify Python Version

Run:

```
python --version
```

Expected:

```
Python 3.11.x
```

Step 10 — Launch Jupyter Notebook

Run:

```
jupyter notebook
```

Browser opens automatically.

Or open:

```
http://localhost:8888/tree
```

Step 11 — Create First Notebook

Inside browser:

- Click New
 - Select Python 3
-

Step 12 — Test Installation

Run:

```
import pandas as pd
import numpy as np

print("Setup Successful")
```

Step 13 — Open Jupyter Later

Every time:

```
conda activate ml_env
jupyter notebook
```

Recommended Folder Structure

Create:

DataAnalysis/

Inside:

- Projects/
 - Datasets/
 - Practice/
 - Notes/
-

Common Beginner Mistakes

Avoid:

- Installing many Python versions
- Forgetting environment activation
- Installing libraries globally

Always activate environment first.

Future Learning Path

Python

↓
Pandas + SQL
↓
EDA + Visualization
↓
Machine Learning
↓
Deep Learning
↓
AI Engineering
